CSE499B – Section 15 – Group 03
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CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 2 (Sprint 2: Plate Reading and Parking Slots)

I. Sprint Overview

This update covers 14–28 October 2026. On the AI side, the second stage of Bangla licence plate recognition was built: a plate reading dataset was assembled, and two approaches were trained and tested, a YOLO character detector and the vision-language reader of [3]. On the backend, parking slots became first-class records with pricing and discount tables, and lots now pass through admin approval. The web portal and the driver app moved further onto live API data.

II. Backend Progress

The database and API were extended so that each parking lot holds individual, bookable slots with flexible pricing.

- **Slot model:** each slot now stores its label, zone and floor, slot type (standard, compact, large, motorcycle, disabled, EV, bicycle), allowed vehicle types, size limits, features and EV charger power, its position in the camera frame (box and polygon), allowed rental types (hourly, daily, monthly), status (active, inactive, maintenance, blocked), access note, an assigned driver for monthly renters, and a unique QR code for check-in.
- **Pricing tables:** `rate_plans` hold hourly, daily, monthly, or flat prices for a whole lot, a zone, or a single slot (the most specific plan applies), with minimum and maximum duration, grace period, overstay rate, deposit, cancellation policy, and optional time windows for peak or night rates. `discounts` hold percentage or fixed discounts with the same scoping, optional promo codes, validity dates, and usage limits. Bookings gained rent type, end time, extension, vehicle plate, discount, and payment status.
- **Slot API:** create one slot or many at once (e.g. `A-1` to `A-50` with shared defaults), list with filters (zone, floor, type, status, vehicle type, and free in a given period), edit one or many slots, and delete. Owners see every field; drivers see only active, bookable slots and public fields. The number of slots is limited by the lot's capacity, and slots with current or upcoming bookings cannot be deleted.
- **Lot approval:** admins list pending lots and approve or suspend them; only approved lots can be booked.
- **Booking history:** drivers can remove past or cancelled bookings from their history (soft delete); active bookings cannot be removed.
- All new endpoints were tested against a local server (46 checks).

III. Plate Reading Dataset

The reader takes a cropped plate and outputs its text. Character boxes from two sources were merged into one YOLO dataset with 103 classes: 10 digits, 26 series letters, 3 words (*Metro* and the short forms used on Chattogram and Rajshahi metro plates), and 64 district names. Each class carries its Bangla text, so the detections can be turned into plate text such as *Dhaka Metro Ga 12-3456* in Bangla script. Letter classes whose names differ only in capitals (e.g. `Da` and `DA`, two different letters) were checked visually before merging.

BLPD contains 2,001 collage images that show four plates each, with the digits blanked out and not labelled; these were removed, together with 13 crops that contain more than one plate. Training crops that duplicate validation or test crops were removed by perceptual hash. Table I shows the result; the test split of [1] is unchanged.

TABLE I
Plate reading dataset (crops per split)

Source	Train	Val	Test
Hossain et al. [1]	3,455	359	218
BLPD [2]	2,839	1,089	560
Total	**6,294**	**1,448**	**778**

IV. Plate Reader Training

YOLO character reader. YOLO26n and YOLO26s were fine-tuned from COCO weights for 150 epochs without early stopping, at image size 320 with batch 64. Horizontal flipping (a mirrored character is a different character) and mosaic (the reader always sees one plate) were turned off; other settings are the Ultralytics defaults. The plate text is built from the detections: place names, *Metro* and the series letter form the top line, the digits the bottom line.

Paper reproduction. The reader of [3] was rebuilt as stated: a ViT-Base encoder with a Transformer decoder and the BanglaBERT tokenizer, AdamW with learning rate 5×10^{-5}, batch 6, mixed precision, 7,800 iterations, and beam search of width 3. Unstated details are our choice: the encoder `vit-base-patch16-224-in21k`, a decoder initialised from BanglaBERT [4], and a maximum of 32 tokens. The paper's MMHQ dataset is no longer available, so both readers use our dataset.

V. Results

TABLE II
Plate reading on the test plates of [1] (218 plates)

Model	Plate acc.	CER	ms
YOLO26n reader	71.1%	5.5%	**1.3**
YOLO26s reader	**75.7%**	**4.4%**	2.5
ViT + BanglaBERT [3]	31.2%	21.2%	43.7

Table II compares the readers on the same 218 test plates. *Plate accuracy* is the share of plates whose full text is correct; *CER* is the character error rate. The YOLO26s reader gets three in four plates fully correct with 4.4% CER and also reaches 0.965 mAP@0.5 over all character classes; its weakest classes are rare letters and districts.

The paper's reader reaches 44.5% CER on our full test split (21.2% on the plates above), against 13.23% reported on MMHQ. On 300 training crops it reaches 7.6% CER, so on our smaller dataset (6,294 training crops against 12,680 in the paper) it memorises plate strings rather than reading them: on new plates the place name and letter are usually right, but the digits are not. The YOLO readers were trained before the BLPD collages were removed; they will be retrained on the final dataset.

VI. Key Findings

Reading each character from its own box is far more reliable on our data than generating the text with a language decoder, and it is about 17 times faster. The YOLO26s reader is the current choice for CityPulse; with the YOLO26n detector, a frame with one plate is processed in about 6 ms.

VII. Frontend Progress

The web portal now runs entirely on live API data, including the authority command center.

- **Data layer:** repositories and hooks for parking spaces, nearby lots, live occupancy, bookings, and device sessions, with an authenticated request helper that refreshes the access token.
- **Maps:** a live map (OpenStreetMap, light and dark tiles) with occupancy pins and clusters, and a location picker for registering lots.
- **Command center:** the home, analytics, enforcement, and zone pages load live lots and slot status, with stress levels derived from occupancy, incidents from AI slot readings, and refresh controls.
- **Owner pages:** lot and camera status badges and a slot grid.
- **Profile:** signed-in device sessions are listed, and profile data from different API responses is normalised.
- **Quality:** unit tests for the insight metrics and profile handling; the API flow and data sources are documented; light is now the default theme.
- **Paper reproduction:** the CNN + GFP-GAN plate reader of [5] was re-run. Its symbol classifier reaches 93.1% test accuracy (paper: 92.4%), but the full pipeline reaches a mean text similarity of 54.3% on 651 test plates (paper: 88% on 30 plates).

VIII. Mobile Progress

The driver app gained lot details, device management, and booking fixes.

- **Parking details:** a detailed lot model with information and extras sections in the details sheet.
- **Active devices:** a screen listing the signed-in devices, with each device's name sent at sign-in.
- **Bookings and profile:** booking details and profile were updated, and slot counts in the booking sheet were fixed.
- **Start-up:** the landing page was removed; the app opens on sign-in.
- **Paper reproduction:** the YOLO-tiny + Haar cascade pipeline of [6] was re-implemented. Two YOLOv8n plate detectors (416 and 640 px) reach 0.988 and 0.987 test mAP@0.5 on 1,153 test images (Table III). The paper's two-stage design with a generic Haar cascade reached only 4.2% F1 (paper: 93%), as the paper's own Bangla cascade is not available.

TABLE III
Plate detection, mobile paper reproduction (1,153 test images)

Pipeline	P	R	F1
YOLOv8n on the full image	93.4%	97.2%	95.3%
Vehicle detector + YOLOv8n	84.2%	83.7%	83.9%
Vehicle detector + Haar	13.8%	2.5%	4.2%

IX. Plan for Next Week

- Retrain the YOLO readers on the final reading dataset and test the full pipeline: frame, plate detector, crop, reader.
- Add plate detection and reading to the AI service as one endpoint, and store plate reads in the backend.
- Backend: endpoints for rate plans, discounts, and price quotes; bookings use the quote and block overlapping bookings.
- Frontend: slot setup and management for owners using the new slot API.
- Mobile: slot selection and rent type in the booking flow.
- Start the RTSP stream worker; continue Dhaka parking footage collection.

References

[1] S. N. Hossain and M. Z. Hassan, "Bangladeshi license plate recognition dataset," Kaggle, 2023.

[2] P. Sarker and S. M. S. Alam, "Cascaded YOLO-based approach for Bangladeshi license plate identification and number recognition," *PLOS ONE*, vol. 21, no. 8, 2026.

[3] N. Hasin *et al.*, "A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR," *arXiv:2603.10267*, 2026.

[4] A. Bhattacharjee *et al.*, "BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla," in *Findings of NAACL*, 2022.

[5] S. Afrin *et al.*, "Bengali license plate recognition: Unveiling clarity with CNN and GFP-GAN," in *Proc. ICCIT*, 2023.

[6] "A cloud-based hybrid model for real-time detection of BRTA-approved licence plates using YOLO tiny and Haar cascade," *arXiv:2609.06507*, 2026.